Distortion caused by controlling transistor implemented in the voltage controlled amplifier

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***Abstract*— This paper describes a voltage controlled amplifier with linearized response to the controlling voltage and discusses its harmonic distortion. As a controlling element, a JFET transistor has been utilized. The design of the amplifier has been made with the aid of Maple mathematical software and National Instruments Multisim circuit simulator. During the simulations, it was found that the controlling transistor would probably introduce significant distortion into the signal path. Therefore a set of measurements on real circuit was made in order to verify that the SPICE model of the transistor provided relevant results during the simulation. Both, the simulated and the really measured parameters of the amplifier are discussed within the framework of this paper.**

Keywords—*Voltage controlled amplifier, Distortion, Harmonics, Linearization*

I. Introduction

Voltage controlled amplifiers are frequently used for various purposes, mainly in the systems intended for analogue signals processing [1, 2, 9]. Among the typical applications, compressors and expanders of dynamics belong. Within this paper, an amplifier constructed for the purposes of audio signal dynamics expander with the expansion rate of 3.3:1 (10 dB) is presented. The aim of the authors was to achieve linear response of the amplification factor to the controlling voltage. With the aid of software simulations, this task was quite easy to fulfill. On the other hand, the methods of simulation failed when minimizing the amplifier's distortion factor. In this paper, the difference between the simulated and measured results is discussed.

The hereby described circuit was intended for operation at the position of hardware dynamics expander. The principle of its operation can be found in [3].

The initial requirements involved low costs and easy-to-be-build functional sample of the amplifier. Therefore, the designers decided to use quad low noise operational amplifier and JFET transistors operating as controlled resistors. Based on recommendations published in [4], local signal feedbacks were employed in order to minimize the circuit's large signals distortion. The linearity of response to the controlling voltage is ensured by employment of an additional reference amplifier, the response of which is mirrored by means of a negative feedback.

However, during the simulations it was shown that for large signals, the total harmonic distortion reaches values that cannot be considered as negligible. This indicated, that the linearization of the controlling element, the unipolar transistor, according to [4], may not be sufficient. This suspicion was subsequently confirmed as soon as the circuit was built and measured.

II. Circuit Description

A block diagram of the circuit is depicted in the Figure 1. Several software tools were used to finish the circuit's design. The calculations of transfer functions and DC biases were made by Maple while the simulation of the circuit's operation was made in Multisim, using SPICE device libraries.

A. General Description

The design of the circuit was processed according to the following requirements:

- Input gain variable from - 10 do +10 dB.
- Nominal input voltage 0 dBu (0.775 V).
- Overexcitation of the amplifier at least 10 dB.
- Frequency response at least 10 Hz to 50 kHz.
- Input gain controlled by a linear potentiometer.

The power supply voltage was set to ± 15 V in order to ensure sufficient voltage swing at the output of the circuit. Provided the maximum amplification factor is 10 dB, the voltage swing of at least 20 V should be possible at the output of the circuit.

The operation principle of the circuit is apparent from the Figure 1. At the front-end of the controlled circuit, there is an input amplifier the gain of which can be adjusted within the range from -10 up to +10 dB. A linear potentiometer was used for better matching in case of multiple parallel channels. However, because of its connection, the gain control law is quasi-logarithmic.

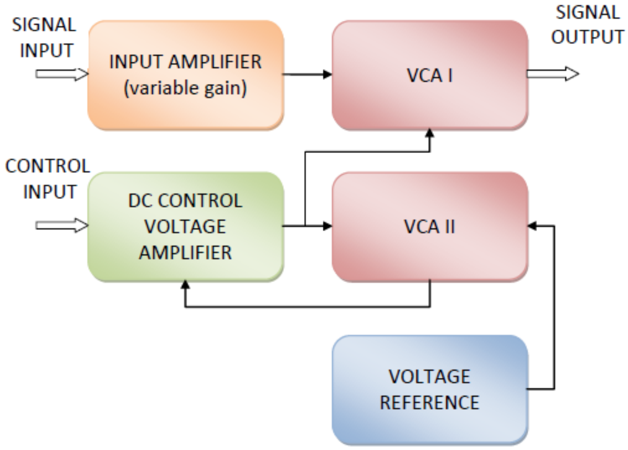

Fig. 1 Block diagram of the VCA circuit

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The main part of the circuit is the voltage controlled amplifier VCA I. Its amplification factor is set by the DC voltage taken from the output of the DC control voltage amplifier. The linear dependence of the amplification factor on the input control voltage is achieved by utilizing the negative feedback from the voltage controlled amplifier VCA II back to the DC control voltage amplifier. A perfect linearization of the response to the controlling voltage is achieved when both amplifiers, the VCA I and VCA II, are identical.

B. Detailed Descritption of the circuit

Circuit diagram of the functional sample that has been designed and constructed is depicted in Fig. 2. Description of the whole circuit exceeds the framework of the paper. Therefore, only the part which is relevant to the occurrence of the harmonic distortion is described here. The more detailed description can be found in [7].

Both VCAs use the same topology. At the positon of the gain controlling elements, field effect transistors T_1 and T_2 can be found. The amplifier VCA I processes the audio signal. It is implemented on the basis of the operational amplifier IC_{2D}. The "reference" amplifier VCA II is used only to create a compensation law for linearizing the response to the controlling voltage. It is based on the operational amplifier IC_{2A} [5]. As the VCAs are identical, the function description below is valid for both of them. Only the device numbers are different.

Both VCAs are based on the topology of a differential amplifier. For IC_{2D}, according to [6], the amplification factor can be simply expressed by (1), when the following assumptions are made: $R_a = R_{33} = R_{34}$ and $R_b = R_{39} = R_{41}$.

$$A_{diff} = \frac{R_a}{R_b} \tag{1}$$

The detailed circuit diagram can be found in the Figure 2. The main controlling transistor is T_1. When it is fully closed, the basic amplification factor of the circuit is defined by the value of the resistor R_{35}. In this case, the amplification factor of the circuit is 0 dB. When the internal resistance of T_1 decreases, the amplification factor rises. Once the transistor is fully open, the resistance of R_{30} defines the maximum amplification factor of 10 dB. According to the parameters of the transistor T_1, neglecting the output impedance of the previous stage and the resistivity of the resistor R_{25}, the gain of this stage can be expressed as follows:

$$A = \frac{R_{41}}{R_{34}} \cdot \left(1 - \frac{\dfrac{R_{35} \cdot \left(\dfrac{r_{DS_0}}{1 - \sqrt{\dfrac{U_{GS}}{U_P}}} + R_{30} \right)}{R_{35} + R_{30} + 1 - \sqrt{\dfrac{U_{GS}}{U_P}}}}{R_9 + \dfrac{R_{35} \cdot \left(\dfrac{r_{DS_0}}{1 - \sqrt{\dfrac{U_{GS}}{U_P}}} + R_{30} \right)}{R_{35} + R_{30} + 1 - \sqrt{\dfrac{U_{GS}}{U_P}}}} \right) \tag{2}$$

The meaning of the variables is obvious. R_9, R_{30}, R_{34}, R_{35} and R_{41} refer to ohmic values of the relevant resistors in circuit diagram in the Fig. 2. U_{GS} refers to the gate-to-source voltage at the field effect transistor in Volts. For the proper function of the circuit it must be lower than 0. U_P is the threshold voltage of the appropriate field effect transistor in Volts and R_{DS0} is the minimal internal resistance of the field effect transistor in Ohms.

As can be seen, the amplification factor is dependent on the value of U_{GS} and it is obvious that this dependency is not linear. The meanings of the quantities displayed at the

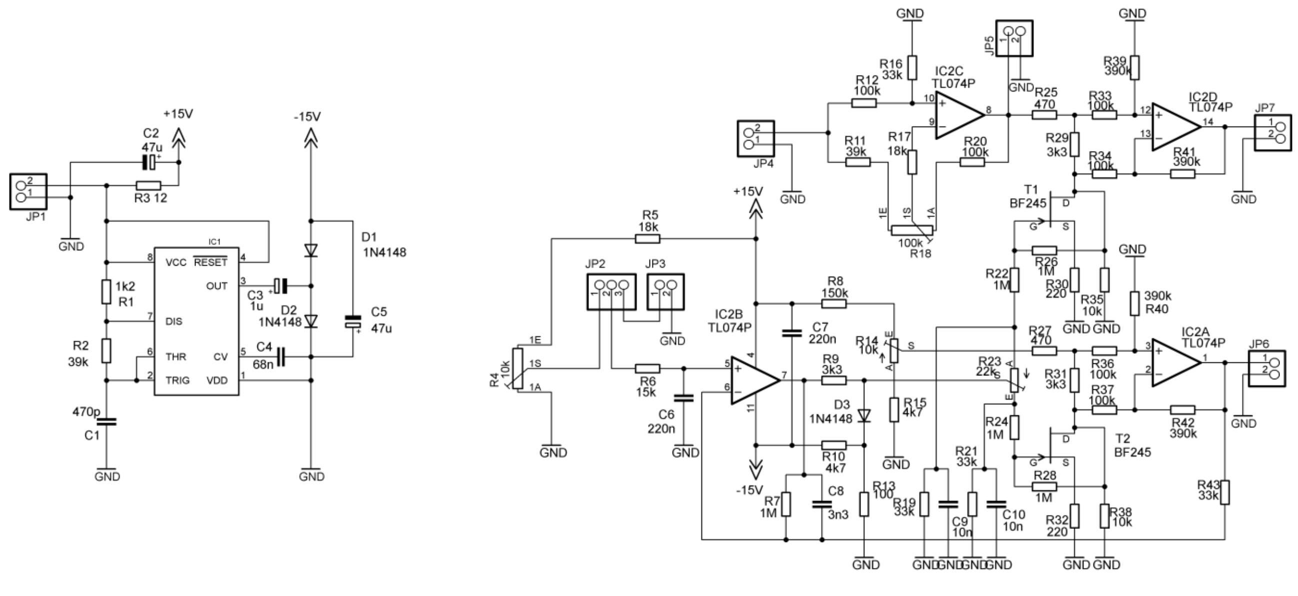

Fig. 2 Detailed diagram of the constructed VCA

appropriate axes are as follows: A − amplification factor [-], U_{GS} − gate-to-source voltage at the appropriate transistor [V], U_P− threshold voltage of the appropriate transistor [V].

As mentioned in the Introduction chapter, there were two main non-linearity problems to be solved. The non-linearity of the dependence of the channel resistance of T1 on the controlling voltage U_{GS} is solved by employment of the additional amplifier VCA II in the feedback of the DC amplifier based on the operational amplifier IC_{2B}. The second problem consists in the fact that signal waveforms processed by the amplifier affect the gate-to-source voltage U_{GS} of T_1 which causes perceptible distortion of the processed signal. The typical N-channel JFET output characteristics are provided in the Figure 3. The asymmetry between the first and the third quadrant of the graph is obvious. This type of the circuit's non-linearity was attempted to be minimized by establishing a signal feedback to the gate of the transistor T_1 in order to compensate the fluctuation of its U_{GS} voltage. This technique is well described in [4]. In the circuit depicted in Fig. 2 this is realized with the resistor R_{26} the value of which was determined by the results of the simulations.

The amplification factor of the VCA I is controlled as follows. The lower boundary of the controlling voltage is defined by the internal reference of 1 V that is created by a voltage divider composed of resistors R_8, R_{14} and R_{15}. The trimming resistor R_{14} allows to fine-tune the voltage. The reference voltage is applied to the input of the complementary amplifier VCA II. Provided the responses of VCA I and VCA II to the controlling voltage are non-linear in the identical way, let us assume that the voltage at the output of VCA II is a function of its controlling voltage multiplied by the non-linearity of the VCA's response. Because the input voltage of the VCA II is the reference as high as 1 V, when the voltage at the output of VCA II is compared to the controlling voltage, the inverse function is obtained. This inverse function allows to cancel the non-linearity of the response to the controlling voltage in case of the amplifier VCA I. The comparison of the controlling functions is processed by the DC amplifier based on IC_{2B} and provided the transistors T_1 and T_2 are identical, the amplification factor of VCA I is linearly proportional to the controlling voltage applied at the input of the DC amplifier. The controlling voltage can either be supplied from an external source through the connector JP_3 or derived internally from the power supply voltage according to the position of the slider of the potentiometer R_4. This depends on the setting of the jumper JP_2. The PN junctions of both transistors T_1 and T_2 is protected by the diode D_3 that does not allow application of positive U_{GS} to their gates. Because small differences in the performance of the transistors T_1 and T_2 were expected, the rotary trimming resistor R_{23} is applied to allow balancing between the responses of VCA I and VCA II that may occur in practice. However, the transistors should be paired and mounted close each to other in order their temperature was similar. Because the setting of R_{23} may affect the function of the signal feedback cancelling the non-linearity of the transistor caused by the processed waveforms, capacitors C_9 and C_{10} were added to cancel this influence for operating frequencies. As the simulations shown that the feedback loop of the controlling mechanism tended to oscillate, the capacitor C_8 was added to limit the controlling loop bandwidth to useful values.

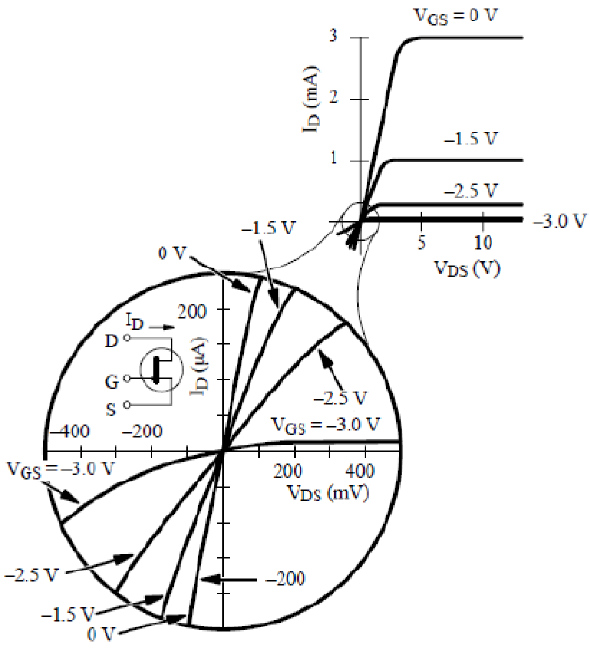

Fig. 3 Detailed diagram of the constructed VCA

III. Observing of the Total Harmonic Distortion

First of all, the distortion was simulated in the National Instruments Multisim software. Afterwards, two different measurements were processed.

A. Simulations

The National Instruments Multisim Software is a widely developed suite for electronic circuits' simulations. It is based on SPICE libraries and algorithms, providing synoptic user interface. This software directly provides Fourier analysis that can be applied to the output of the circuit being driven by a harmonic signal of a constant amplitude and frequency. The output waveform is extracted by means of transient analysis and its accuracy depends on the proper settings of parameter sets for both analyses. Spectrum calculations are performed by means of Discrete Fourier Transformation (DFT). For the calculations, only the second cycle of the fundamental component of a time-domain or transient response (extracted at the output node) is used. The first cycle is discarded for the settling time. The coefficient of each harmonic is calculated from the data gathered in the time domain, from the beginning of the cycle to time point "t". That is set automatically and is a function of the fundamental frequency. This analysis requires a fundamental frequency matching the frequency of the AC source or the lowest common factor of multiple AC sources. A screenshot of the Fourier analysis dialogue, depicting the typical setting, is depicted in Fig. 4. The THD (total harmonics distortion) simulation for the gains of both VCAs of 0, + 3, + 6 and + 9 dB was processed.

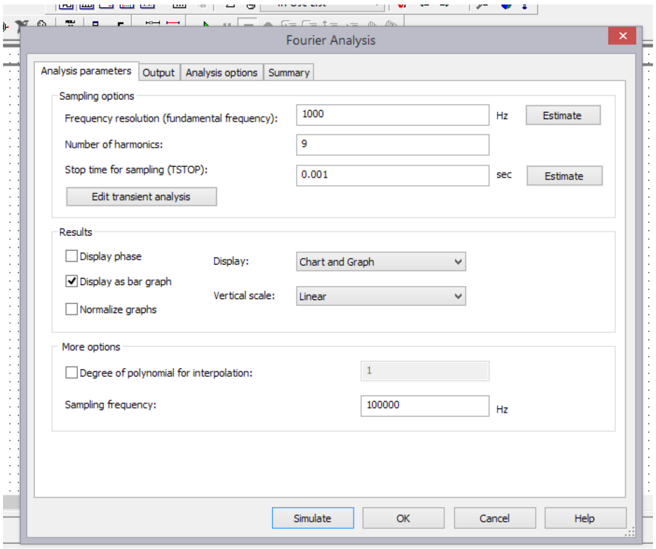

Fig. 4 Setting of Fourier analysis in National Instruments Multisim

During the simulation a sinusoidal signal was used the frequency of which was 1 kHz and the amplitude of which was as high as 0 dBu. The THD was computed by means of Fourier transformation and the signals at the front-and and at the back-end of the VCA I were compared. In all cases the simulation shown THD < 0.01 %, provided the value of the linearizing resistor R_{26} was as high as 1 MΩ.

Therefore, from the point of "simulation" view, the correction according to [4] was properly set.

B. Distortion Measurement with Oscilloscope

Before the measurement was processed, it was checked whether the power supply voltage in both branches was ± 15 V ± 10 % and the adjustment of the trimming resistors R_{14} and R_{23} was processed as follows, enabling linearization of the VCA's response in the following points:

1. The position of the slider of the trimming resistor R_{23} was set to 50 %. The DC voltage at the control input was set to 1V and the position of the slider of the trimming resistor R_{14} was set in order to achieve the VCA's gain as high as 1.
2. The DC control voltage was increased to 2.5 V and the slider of the trimming resistor R_{23} was moved so the VCA's amplification factor was as high as 2.5.
3. The setting of the trimming resistor R_{14} was checked.
4. The steps 2 and 3 were repeated until the linearization was achieved for both points (1 and 2.5 V of input DC control voltage).

After proper setting of the circuit, the measurement procedure was as follows:

1. The input of the circuit was fed with a sinusoidal signal the frequency of which was 1 kHz and the voltage of which was 1V_{RMS}. This refers to a slight overexcitation of + 3 dB.
2. The driving signal was connected to the connector JP_4 and the output signal was measured at the connector JP_7.
3. No load was connected to the connector JP_5.
4. The control voltage was adjusted internally by the potentiometer R_4.
5. The gain of the preamplifier was set to 0 dB.
6. While the level and the frequency of the input signal remained unchanged, the controlling voltage was increased in steps corresponding to the gain of 1, 1.5, 2, 2.5 and 3.
7. For each of the gain setting, the output signal spectrum was monitored by means of an oscilloscope supporting the FFT processing.
8. The levels of 2nd and 3rd harmonics were extracted from the spectrum of the observed signal and their relative levels were calculated.

The results obtained by measurement with the oscilloscope are enlisted in Table 1. This table includes the information on the VCA I amplification factor set by the controlling voltage, the levels of the second and the third harmonics in percents and the total overexcitation of the circuit. In all cases, the preamplifier's gain was set to 0 dB.

The third harmonic distortion is unacceptable, as the value of 8.2 % at the gain of 1.5 is considerably high. This indicates that the linearization of the field effect transistor proposed in [4] would not work properly at all gain settings.

TABLE I. RESULTS OBTAINED BY OSCILLOSCOPE

Gain [-]	**Level of Harmonics [%]**		**Total Over Excitation [dB]**
	2nd	***3rd***	
1	1.0	3.1	+ 3.0
1.5	0.9	8.2	+ 6.5
2	0.7	3.1	+ 9.0
2.5	0.4	1.0	+ 11.0
3	0.1	0.4	+ 12.5

C. Distortion Measurement with Vector Network Analyser

Because the FFT measurement by means of an oscilloscope may be considered more quite indicative than accurate, the authors of this paper decided to repeat the distortion measurement with a "hacked" vector network analyser (VNA).

Usually, the VNA is employed according to Fig. 5. The VNA produces signal that is monitored at the front-end and the back-end of the measured device and the appropriate transfer function is calculated afterwards. For the purposes of this experiment, the setup was changed as depicted in Fig. 6. While the VNA produces and scans signals within the required spectrum, the circuit is fed with a stationary signal produced by the external generator. Subsequently, the spectrum calculated by the VNA does not correspond to the transfer function of the tested circuit, but it fits the response of the tested circuit to a single harmonic frequency at its input quite well, provided the total harmonic distortion of the generator is much lower than the values expected to be measured. For the purpose of this experiment, the total harmonic distortion of the generator was measured and its value reached approximately 0.05 %.

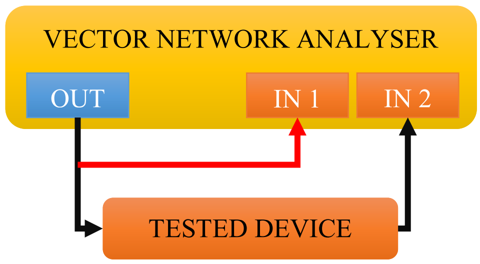

Fig. 5 Typical application of VNA

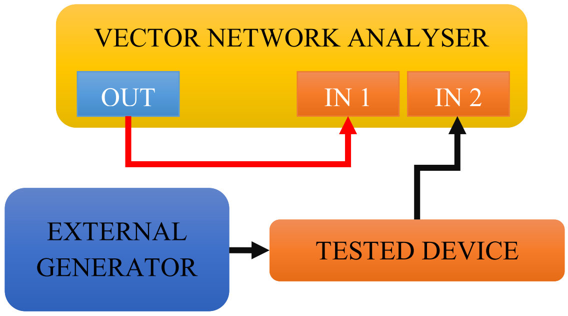

Fig. 6 Hacked application of VNA for measurement of levels of harmonics

In this measurement, all the steps mentioned in the previous subchapter were repeated, but the output of the circuit was monitored by means of the VNA instead of the oscilloscope.

The measurements were made for two different input levels, 1 V_{PK} referring to 0 dBu and 0.35 V_{PK} (- 12 dBu).

An example of a spectrum measured at the worst case, where the amplification factor was A = 1.5 and the input amplitude was 1 V_{PK} is depicted in Fig. 7. The level of the 3rd harmonics is – 28.3 dB compared to the 1st harmonics, which corresponds with the distortion of 3.8 %. Although this is a better result than the rough outputs obtained by the oscilloscope, the worst case distortion level is still quite high, although more accurate method was used for its evaluation and the input level was decreased from + 3 dBu to 0 dBu.

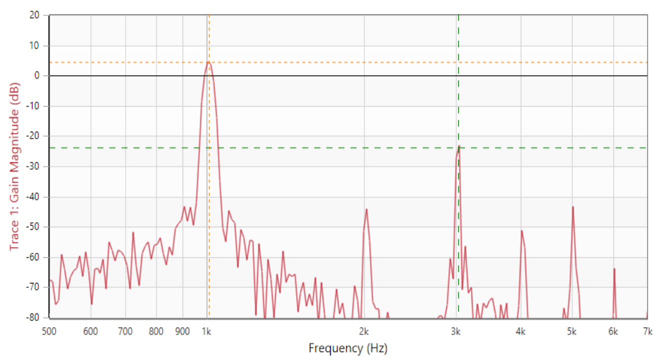

Fig. 7 Output spectrum of the VCA when driven by harmonic signal with the frequency of 1 kHz and amplitude of 1 V, while the amplification factor is set to 1.5.

The results obtained by this measurement are enlisted in the Table II (input level 0 dBu) and the Table III (input level – 12 dBu).

TABLE II. RESULTS OBTAINED BY VNA, INPUT LEVEL 0 DBU

Gain [-]	Level of Harmonics [%]		Total Excitation [dB]
	2nd	*3rd*	
1	0.20	0.12	0
1.5	0.50	**3.80**	+ 3.5
2	0.13	1.20	+ 6.0
2.5	0.20	1.00	+ 7.9
3	0.40	0.15	+ 9.5

TABLE III. RESULTS OBTAINED BY VNA, INPUT LEVEL – 12 DBU

Gain [-]	Level of Harmonics [%]		Total Excitation [dB]
	2nd	*3rd*	
1	0.25	0.28	- 12.0
1.5	$\leq$ 0.10	**1.60**	-7.5
2	$\leq$ 0.10	0.65	- 6.0
2.5	0.16	0.25	- 4.1
3	0.18	0.26	- 2.5

IV. CONCLUSIONS

This paper describes a design and construction of the voltage controlled amplifier with linearized response to the controlling voltage, which should be suitable for application in audio compressors or expanders.

The design of the circuit was made according to the JFET transistors application note [4]. The aid of Maple computing software and National Instruments Multisim circuit simulator based on SPICE was utilized. During the simulations it seemed that linearization of the controlling transistor T_1 by the resistor R_{26} is sufficient. However, during the measurements it was found that further improvements must be made in order to use the circuit for audio signals.

The first measurement on the circuit was simply made by means of the oscilloscope with FFT calculation with the circuit being slightly overexcitated. As the simulations provided THD values as low as approximately 0.01 %, it was expected to achieve real distortion levels below 1 %. However, during this measurement it was shown that 3rd harmonics reaches the level of approximately 8.2 % when the amplification of the circuit is set to 1.5. The deviation from the expected values was so great that a decision to check the distortion by a more accurate way was made.

The second measurement was processed with the help of the vector network analyser Bode 100. It screened the output spectrum of the measured circuit in the frequency range from 500 Hz up to 7 kHz while the input of the circuit was driven from a separate signal generator. As a result, a set of spectra was obtained for different input levels and amplification factors set by the controlling voltage. These results are enlisted in the Table 2 and Table 3. This measurement shown that even with the input level decreased to 0 dBu or – 12 dBu respectively, when the amplification factor of the circuit is set to 1.5 (+ 3.5 dB), the 3rd harmonics reaches quite large levels of up to 3.8 %.

This experiment shown that simple linearization, as described in [4], cannot cancel the 3rd harmonics distortion in the whole operating range of the transistor as the levels of intermodulation changes with the amplification factor of the circuit. As the distortion decreases with the excitation level, the improvement can be reached when the level of the processed signal is decreased significantly. On the other hand, this measure would result in worsening of the signal-to-noise ratio. Therefore, other ways are currently being tried in order to improve the linearity of the circuit.

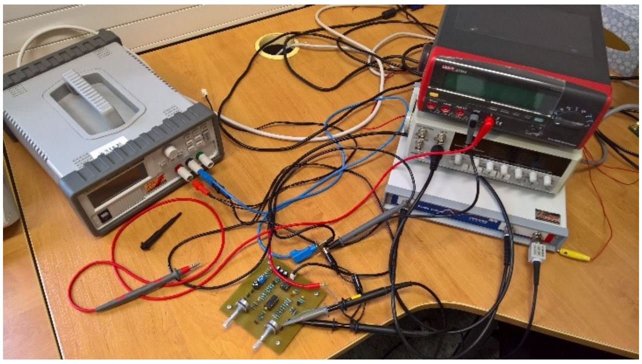

Fig. 7 Testbench with the VNA

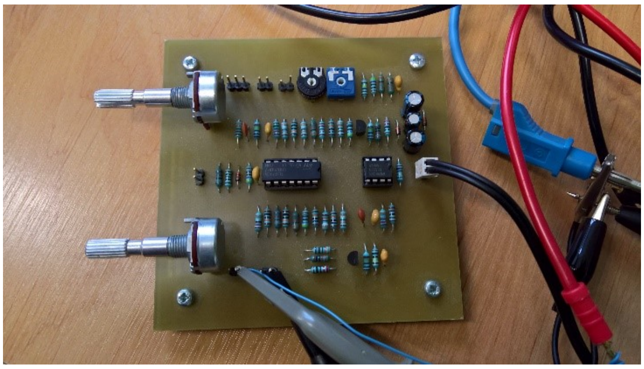

Fig. 8 Detail of the VCA

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